

Module 9: Capital Budgeting Techniques

Topic 163: Capital Budgeting-1

In Capital Budgeting, we talk about **such huge amounts which we invest in our business for longer period**, for long term. It means that these amounts are too big and also their time period is also longer.

Because of huge and large amounts and longer time period, the risk level is very high. So, it is very important for every business to plan all those investments, all those amounts which are to be put into investment for longer period of time very carefully.

Capital Budgeting:

The topic of Capital Budgeting is actually related to the purchases of large assets meaning that such fixed assets which are of bigger amount i.e., plant, machinery, equipment etc. in which a lot of cash outflow will be needed and this amount is invested for many years.

So, when we are planning to see that what amount is required for the purchase of new assets or fixed assets or we can say that we want to start a new factory or new production plant so, this is called the **Capital Expenditure**.

And planning these Capital Expenditures is called **Capital Budgeting**.

Whenever we make a plan for the future activities, it is called a **Budget** where the plan contains the figures, the amounts in rupees and dollars or in some monetary terms.

Concept of Capital Budgeting:

Whenever we are planning to invest in a new plant or machinery or any new production line where a huge amount of money will be invested, we need to do some calculations with reference to how much amount of money will be required for that investment and what will be the return from that investment and what time period will be required to get those returns on our investment. Then we have

certain benchmarks or levels with which we compare the findings or the results of our calculations and we make a decision.

In simple words, we can say that if we want to start a new project and we want to make decision that if we have to invest money in this project or not then **Capital Budgeting helps us to make this decision.** Obviously that decision will be based on the view point whether it will be giving us enough return or profit that we anticipate from this project.

Significance of Capital Budgeting:

The significance of Capital Budgeting is huge as we have said that there is a huge amount involved in it. If we get wrong about any small expense then its significance is very small or its impact on business is very less and we use a term for this kind of expense that is called **Immaterial** meaning that there has not been effect on our business by doing such small wrong expense.

But if we are making our wrong decision about a huge sum of money that we are putting in some project and it becomes a failure so, a lot is lost and the business might have to discontinue.

Techniques of Capital Budgeting:

So, it is extremely important that how do we perform this process of Capital Budgeting. We have some Capital Budgeting techniques and we will see these techniques one by one i.e., **Internal Rate of Return, Net Present Value, Payback Period, Profitability Index.** These are such techniques that we use in the process of Capital Budgeting.

Criteria of Capital Budgeting:

There is a criteria of decision making in Capital Budgeting where **criteria means that how we will make decision that one project is acceptable for us or not.** Or in other words, if we are financial experts and we have advice the management that whether the project should be considered for investment or not then how would that decision be made. What are that criteria of selection/rejection or acceptance/rejection on the basis of which we will make decision.

So, these criteria are also related to the techniques that we will be talking about that we name them **Discounted Cash flow methods** that we will discuss in detail in next lectures.

On the basis of **Discounted Cash Flow methods** or such techniques, we have built criterion related to every method which will tell us that result in such a way that we should accept or not.

Cashflows of Capital Budgeting:

The whole process of Capital Budgeting is related to Cashflows meaning that this is not related to profit but Capital Budgeting's whole calculation is based on the Cashflows and this is something which is very important to understand.

Cashflows are actually the actual amounts of money received or paid but profit is a difference of revenue and expenditures. There is a possibility to manipulate the amount of profit in any given year but there is no possibility to manipulate the amount of cash.

So, we need to remember that all these Cashflow techniques are based on the actual Cashflows that we will have. So, when we do the process of Capital Budgeting in which we will anticipate the Cashflows. And these are the same Cashflows as we have discussed earlier that how much will be cash inflows and how much will be cash outflows.

We can conclude our discussion that **Capital Budgeting** is extremely important because this is about long term investment and huge amounts are involved and to do Capital Budgeting we utilize cash inflows, we project these cash inflows and then apply certain techniques which are called **Discounted Cashflow methods** in order to decide about the accepting or rejecting a certain proposal of a new project.

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Topic 164: Capital Budgeting-2

In this lecture, we will talk about Capital Budgeting Techniques and we will discuss that these are actually called the Discounted Cashflow Methods.

These techniques or Discounted Cashflow Methods are applied to make a decision about accepting or rejecting a certain investment proposal or a certain project in which we want to put our money.

Capital Budgeting Techniques:

As we said that we have used the term Discounted Cashflow Methods where the term Discounted means to calculate the value. When we use it for any amount, we have discounted it meaning that we have to calculate its present value. And the Cashflow comprises of cash inflow and cash outflow.

So, the term Discounted Cashflow means a method or a technique in which Cashflows' present values are calculated and then we make a decision about accepting or rejecting a certain project of investment.

These techniques are given below:

Payback Period:

This method initially in its raw form or crud form is not a Discounted Cashflow method because in it we do our calculations without calculating its present values but this is a payback period which we can apply by converting into Discounted values meaning that it is not just that this method is applied without considering present value, it may be used by considering the present values.

What does this Payback Period method tell us?

It simply states that in how many years we will be able to recover our original investment that we are putting in certain project. So, whether we will be able to get our original money back in 5 years or 3 or 10 years and this time period is called the Payback Period.

This method does not tell us more than this that in how many years our original investment will be returned back or paid back. And we should have in our mind before putting our money in any project that in how many years we want to get this money back (i.e., should our invested by back in 5 years or 7 years) so, we have a certain benchmark which is also taken from similar kind of business that if there were a project and in how many years we got our investment.

Net Present Value:

Present value is that if there is a future amount and we discount it by applying the formula and calculating its present value (PV is the calculated by subtracting interest from the future value) and we apply this method or technique on the basis of present value therefore this method is call the **Net Present Value method**. Here **netting off** means that when we deduct the outflows from our inflows after calculating the present value then what we do get is the **Net Present Value**.

In this method, we do consider time value of money, we do consider the present values so, this is certainly the **Discounted Cashflow method**.

Internal Rate of Return:

In this method, we calculate that rate of any project which this project will give us by earning during useful life. So, this method helps us in determining the actual rate of return of that project which is again based on the anticipated Cashflows which we calculate. On the basis of this, once we calculate that rate of return of the project, we compare it with the required rate of return.

It means that whenever any business or company goes for any project then it has a rate of return as benchmark which is the opportunity cost.

It means that it wants to earn how much percent of rate and once we calculate the Internal Rate of Return of that project which is the rate that project will actually earn for us and we compare it with the required rate of return.

If both the Internal Rate of Return and Required Rate of Return will become equal that its good for project and we will accept that project. And if our required rate is less but project is giving us more rate then its wonderful for the project but if our

required rate is more and project is giving us less than that then we will not accept that project.

Profitability Index:

The Profitability Index is a variation of Net Present Value but Net Present Value method is in absolute terms and Profitability Index is a relative measure.

We can conclude our discussion that there are four commonly used Capital Budgeting Techniques which are also called Discounted Cashflow methods; first one (**Payback Period**) may be applied without discounting or with discounting but the other three methods (**Net Present Value, Internal Rate of Return, Profitability Index**) are certainly applied by considering the present values or the discounted values.

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Topic 165: Capital Budgeting-3

In **Cash Budgeting**, the decisions are made on the basis of Cashflows and not one the basis of profit. We have discussed that Cashflows are more authentic or more reliable in terms of their absoluteness meaning that profit can be manipulated and profit can be increased or decrease in a certain period of time but Cashflows are something which is more concrete as we cannot increase or decrease Cashflow amount because this is the sum of money which we are paying or which we are receiving.

And all these Cashflows Techniques, all those Discounted Cashflow methods (Payback period, Net Present Value, Internal Rate of Return, Profitability Index) are based on the Cashflows meaning that we can draw the results of acceptance or rejection by using these Cashflows.

So, it is extremely important for us to estimate these Cashflows and we have to be very careful about estimating the future Cashflows related to a project. The **Capital Budgeting is based on that in the project where we are going to invest** then what amount of cash we have to invest in this project meaning that what will be our

outflow and when project or its operation become started then what will be our cash inflows. These cash inflow are also estimated and these are anticipated Cashflows inflow and these are not actual Cashflows because we are referring to the future so, we have to be very careful about estimating this cash inflows and then at the end of the project when once project will be terminated then that time what will the position of our Cashflows.

So, decision will be made by applying the Discounted Cashflow Methods are authentic and reliable.

Estimating Cash Flows:

To estimate these Cashflows first we have to consider that what are these Cashflows.

Initial Cash Outflow:

When we actually invest in a project if it is a big machine or equipment that we have purchased then there will be cash outflow. When we are starting a new factory or project or a plant then the Initial Cash Outflow is huge and at many times this is not at one point of time meaning that it is not such that we outflow all that cash at one time but outflow happens within segments.

But for the purpose of learning the cash capital budgeting techniques, we assume in start that all the cash outflow has happens at one time but then we can expand or make little more complicated by making different cash outflows at different points of time.

So, the **Initial Cash Outflow** is the price that we are paying for purchasing that new plant or equipment and the second thing is that we have to consider the expenses or cost that we have to incur on bringing the project on which we have to pay government duties and others taxes and then we have to install that plant or equipment. All these outflows related to that plant or equipment or asset will be part of our Initial Cash Outflows.

Incremental Cash Inflows:

Once we have incurred that expenditure which we have made an outflow then the second thing is that it will start its operations meaning that we will start to use or operate that plant or asset.

As a result, sales or revenue will be generated and that revenue would results into cash inflow but we know that the whole amount of sales or revenue is not the net cash inflow. We have to subtract that amount which we have to pay in the form of expenses or cost from the figure of revenue or sales. So, when our plant or asset get started to utilize then we get cash inflows over its useful life which are called **Incremental Cash Inflows.**

This **Incremental Cash Inflows** is actually the difference of the inflows coming from our sales or revenue and cash outflow in the form of cost or expense and we also adjust the non-cash expenses to calculate it.

Terminal Cashflow:

The Cashflow of the last year of that project which is called the **Terminal Cashflow**. Here **Terminal** means where our project concludes and that year how much will be our cash flow. It is usually cash inflow meaning that our incremental cash inflow will be that year also.

And after that when we terminate that project then it may be that we sell this project or asset or plant which has been completed its useful life and we may also get some inflow from that sale or may involve some taxes or charges then all these things are calculated in the last year's calculations which is different from the Incremental Cashflow.

We can conclude our discussion of Cashflows that it is important that if Cashflow is rightly calculated then on the basis of it, our Discounted Cashflow methods will be applied rightly. And the results of those methods then effectively used for accepting or rejecting a project. And these Cashflows are of three types; outflow which is called the Initial cash outflow when we are investing our money in the project, Incremental cash inflows during its useful life and the last one is Terminal cashflow which is in the last year of the useful life of that project.

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Topic 166: Capital Budgeting-4

We will be discussing one very specific important element which is very important in the calculations of Cash Budgeting and Cashflows that is **Depreciation and the Methods of Depreciation and the Treatment of Depreciation in our Cashflows.**

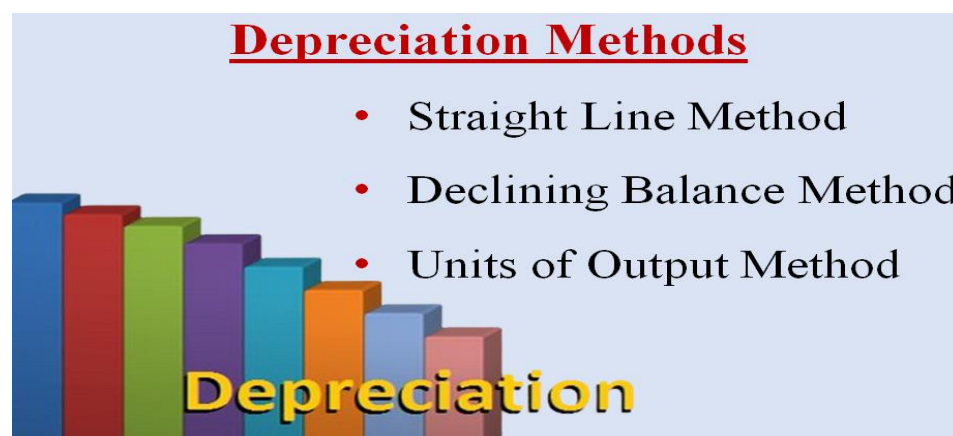
We know **Depreciation** as a non-cash expense meaning that such expense for which we have not to pay cash which this expense is recorded and also we don't have to pay any cash against that expense after that.

So, the term **non-cash expense** refers to not the payment of cash at the time of expense or in the future because Depreciation does not require a payment of cash at the time of expense or in the future so we call it a Non-Cash Expense.

There are different methods of applying Depreciation or calculating the amount of Depreciation. Because all these methods actually give us different amounts of Depreciation so, if the amount of Depreciation is different, the amount of profit will be different therefore it makes a difference in the amount of calculations of profit.

Depreciation Methods:

When we talk about the commonly used Depreciation methods of accounting, then we will refer to three of these methods one by one which are given below:



Straight Line Method:

In this method of Depreciation, we divide the cost of an asset on the whole life of that asset, every year we charge the same percentage of depreciation.

Declining Balance Method:

In this method of Depreciation, we have more depreciation in the start and in coming days it is becoming less because we apply the rate of depreciation on its book value and as book value reduces, the amount of depreciation also reduces.

Units of Output Method:

This method of depreciation is little less systematic as compare to the first two methods (Straight line method and Declining Balance Method). It means that in these two methods, there is a formula which tells us about the calculations of the cost allocation or depreciation every year but Units of Output method is such a method which is based on the time for which an asset is actually used or how much is it used and how much units are produced from this. Then it may be that sometimes depreciation may be more or sometimes less.

How to treat Capital Budgeting?

Our actually focus of discussion is not to calculate depreciation and see its methods how they are applied but our focus is that what is the significance of Depreciation in our Cashflows and how we see our depreciation methods in the Capital Budgeting because **Depreciation is a non-cash expense** so, being a non-cash expense, it does not disturb our Cashflows and it does not affect our Cashflows.

But practically when we are calculating the amount of cash inflow which we named **Incremental cash inflow** where we said that we subtract our expenses from the sales and then we get a figure of profit which we consider it in the form of Incremental Cash inflow.

But this will be equal to the Incremental Cash Inflow when we have adjusted the non-cash items or depreciation in our profit meaning that when we finally calculate our profit then we again add the amount of depreciation in the profit because there

has no cash outflow from that amount. We will see here that how and which place the depreciation is treated in our incremental cash flow and how we adjust our incremental cash inflows.

We can conclude our discussion that as Depreciation is a non-cash expense therefore is representative of all non-cash expenses and as depreciation will be treated and other non-cash expenses will be treated in the same way.

Non-cash expenses are actually those expenses which are subtracted from our Revenue in the Profit & Loss but there is not cash outflows from them. Therefore when we determine the amount of incremental cash inflow then we add that amount of depreciation or non-cash expense in the final amount of profit and from which we will get the correct amount of cash inflow or incremental cash inflow.

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Topic 167: Capital Budgeting-5

In Capital Budgeting, we have seen that it is something which is related to investing a lot of money in purchasing a new fixed asset or a new project. So, when we want to purchase a fixed asset or we want to start a new project or even we want to replace an existing asset with another one, we require a lot of investment.

So, that is why, we need to look into the calculations and decide whether this investment will be profitable or not.

Initial Cash Outflow

We use the Cashflow for the investment that we are going to make in fixed assets and do not consider the profit. So, estimating Cashflows is the essential and key part of financial decisions related to Capital Budgeting. In this, we have got cash inflows and cash outflows so, whenever a company or business would make an investment in an asset or in a project that would require a cash outflow.

Initial Cash Outflow



After that, when that project get started or starts its operations and to give us profit then that time cash inflows have started to come in our business. It means that in Capital Budgeting, we have got outflows and inflows and obviously these cash inflows and cash outflows are spread over a certain period of time in the future. Because the time at which we are making the decision is before making the investment in the project. So, that is why **it is called the Budgeting or planning.**

Firstly, we have to see that what will be our cash outflow for the investment in any project. Here another important point that we need to remember that for the purpose of understanding the concept of Capital Budgeting, we make some assumptions with reference to the cash flows.

The first assumption that we take is that our all cash outflow has been happened at the time of investment in any project or at zero point or before the start of the project. Whereas in reality that outflow is not at one point of time but it is after several months or days that we have to keep on making the payments which is the outflows.

Likewise, when we talk about inflows and again to understand the working and calculations of Capital Budgeting, we also assume that the inflows are also at the end of a particular year. So, by keeping in mind this assumption that we will try to know about some things about cash inflows and cash outflows.

Purchase Price:

Cash outflow is happened at the start of the project or during the purchase of asset. It is obviously the purchase price which is the most significant amount of investment.

This purchase price is of three types as it is quoted by the vendors or suppliers.

1. Purchase Price

- a. FOB
- b. C & F
- c. CIF



FOB (Free on Board): In some of the books, we will find it as Freight on Board this means that this price only includes the purchase price of that asset and no additional cost is made a part of them.

C & F (Cost & Freight): If the vendor or the supplier is also including the Freight which will be paid on bringing that asset from the supplier premises to our premises then this price is called C & F price which means cost and freight.

CIF (Cost Insurance & Freight): At times, the cost of insurance is also added in the purchase price meaning that it includes cost and also includes the insurance expense which will be spent in bringing an asset to our factory or premises and obviously the Freight. And we call it CIF (Cost Insurance & Freight).

Direct Costs:

There are some other costs which we call Direct Costs and these costs are also made a part of our investment. If we make any expense or cost a part of any asset

then it is called **Capitalized Cost**. So, the costs which are capitalized and made a part of asset are the **Direct Costs**.

These Direct Costs are given below:

2. Direct Costs

- a. Freight
- b. Insurance
- c. Installation
- d. Setup
- e. Duties & Taxes



Freight: The Freight or fare that we pay for purchasing an asset and bringing it to our premises or factory.

Insurance: We have to pay Insurance expense so that we do not have to face any loss during the project.

Installation: When an asset is received if it is a plant or equipment or machinery then it has to be installed. So, the cost of installation in case of big projects there is a huge installation costs and it takes a lot of time to install that and this installation cost is also part of our investment.

Setup: Once an asset has been installed, we have to start operating that asset and bring it into that condition where the routine normal production can be started, is called the **Setup Cost** which is also part of our Direct Cost.

Duties & Taxes: Along with Setup cost, there are Duties & Taxes cost. In every country, when an asset is imported or it is purchased, there are some duties of government and there are certain taxes which are not adjustable, all these un-adjustable taxes and the government duties are also the **Direct Expenses or Direct Costs** and these are also Capitalized and included in the Initial Cash outflow of the asset.

So, collectively all these costs in addition to the purchase price of the asset is the Initial Cash Outflow of the asset which is then to be matched with the inflows.

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Topic 168: Capital Budgeting-6

We are continuing our discussion on **Capital Budgeting**. Before this lectures, we have been talking about the Cashflows and why Cashflows are taken for the decisions related to the investments and capital assets. We have talked that we make decisions about Cash Budgeting by considering the Cashflows and we consider cash inflow and cash outflow in place of profit.

We talked about the first part of our estimated Cashflows and that was a Cash Outflow and we said that when any new asset will be purchased or any project will get started and all the cost which will be incurred in order to purchase or install or bring it to the factory, will be Capitalized or added and that is made up the part of total value of our investment. And this comprises of all these items and this is the Initial cash outflow.

Incremental
Cashflows



The point to remember here is that it is estimated cash outflow and all these cash flows that we have talked about with reference to the outflow and then we will talk about with reference to the cash inflows, all are the estimated Cashflows.

If that estimate of our Cashflows will not be accurate then our Capital Budgeting decisions would not be very good or correct. So, it is extremely important when we are calculating these cash inflows and outflows, we are very realistic and we consider all the possible elements while making the calculations of these Cashflows.

Once we have made the investment or when we have purchased an asset, have installed and have setup then this project or asset have started to operate and once the operations starts and the production starts then we get the product. The product is then sold in the market and as a result we get Revenue after deducting the expenses and cost of productions and marketing and all those operations will come up with the figure or amount of profit which is the **net profit**.

In this calculation we also have to consider some of the expenses which may not require cash outflow meaning that generally we say that all the expenses require cash outflow or when we have to incur an expense either we will pay cash for it or will pay in the coming time or have to pay in advance.

But there are some items which actually do not require cash outflow so, these **Incremental Cash Inflows** (it means that during the useful life of the asset, the inflows that we will receive from that project that is called Incremental Cash Inflow) are calculated by considering the revenue, cost and expenses of operations and also the non-cash expenses.

Incremental Cash inflow calculations:

Incremental Cash Inflows		
a		Additional Revenue
b	-	Additional Depreciation
c	=	Net Change in Profit before Tax
d	-	Additional Income Tax
e	=	Net Change in Profit after Tax
f	+	Additional Depreciation
g	=	Incremental Cash Inflow

We can conclude our discussion that **if we want to calculate the Incremental Cash inflow of any asset then for that we have to see Additional Revenue**, then have to deduct depreciation in order to calculate the figure of profit on which we have to pay tax and then after deducting the tax from that profit then we have to add back that depreciation amount and then we will get the figure of Incremental Cash Inflow.

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Topic 169: Capital Budgeting-7

In previous lecture, we have talked about Incremental Cashflow and it is actually that cash inflow which we get through any project or whenever a project starts its operations and we generate some production and then sale in the market and as a result we get revenue. We also have to incur operational cost to earn that revenue meaning that we have to pay expenses, labor and other overheads but we also get some Additional amount of revenue with all this which is ultimately translated in the profit.

Incremental Cashflows-2



We have discussed in previous lectures that Profit and Cashflows are not equal so, Incremental Cash Inflow is not equal to the amount of profit. Because of the non-cash items meaning that there are some such expenses or costs which are actually considered as cost or expense and these are subtracted from the revenue in order to calculate the figure of profit but there has no cash outflow happens from these expenses or costs (non-cash expense or cost).

So, we call them non-cash items and the example or basic non-cash item which is almost in every capital budgeting decision is the Depreciation of the asset. By definition, the Depreciation of an asset is the systematic allocation of the cost of the asset over its useful life.

So, the Incremental Cash Inflows that we get in which one expense is the expense of depreciation that is non-cash expense and at first stage we subtract this depreciation from our revenue in order to get the figure of profit on which we have

to calculate the tax. But in after tax profit, we add back that same depreciation amount in this profit after tax to get the figure of **Incremental Cashflow.**

Why deduct depreciation?

If we do not deduct depreciation then our profit will be overstated meaning that as depreciation is non-cash expense (which can be subtracted from revenue for the purpose of calculating the income tax) we firstly deduct depreciation in order to reduce net profit and save the payment of our income tax.

So, we deduct that depreciation amount and get lesser amount of profit and hence a lesser amount of tax.

Why add back depreciation?

Once we calculate the profit after tax then to calculate the amount of Cashflow, we add back that expense (from which no cash outflow happens but expense occurs) in our final profit after tax in order to calculate the correct figure of cash inflow.

And that is why the profit is different from the Cashflow meaning that the difference between both profit and Cashflow is that all cash and non-cash expenses or costs are subtracted from the amount of profit but **in Incremental Cash inflow,** we only subtract all the expenses or costs from the profit except non-cash expenses and in other words we add back that amount of expense and nullify the effect of it.

Other Non-cash expenses:

In addition to depreciation, there are some other non-cash expenses which can be considered e.g.,

Amortization: The word of Amortization is used from difference references in financial management but a common use of **Amortization** is that it is a decrease in the value of intangible assets I.e., we name Amortization to the depreciation or decrease in value or allocation of cost of the patents, copyrights etc. And it is also a non-cash expense.

Depletion: Such assets which are consumed by itself i.e., if we have got some mines, some land which is consumed itself so, in that case, we use the term Depletion for the decrease in value or allocation of cost.

Bad Debts: The Bad Debts can also be an example of **non-cash expense**.

We can conclude our discussion that **Incremental Cash inflow** is actually the name of the calculation of the inflows that we will get from our project and investment during the useful life of any project. And this is something we have to do on estimated basis and these cash inflows are certainly more than the cash outflow meaning that the outflow which we have to do in the start and then the cash inflows (that we are calling Incremental cash inflows) are certainly more than that cash outflow.

When the useful life of that asset will be completed then the cash inflow will also be stopped. One thing that we need to keep in mind that Initial Cash outflow is at zero point or at the start of the project or at the time of making investment but Incremental Cash inflows are in the future at the end of year one, two, three so on.

And if its useful life is 10 years then over the period of 10 years we will be getting these cash inflow and that is why these are called Incremental Cash inflows.

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Topic 170: Terminal Cash flow

The word **Terminal** means ending Cashflow or Cashflow at the end of the useful life of the project or the asset. Whenever we do capital investment or start a new project or purchase a new asset then it has a useful life and during this useful life, it can be productively be used in our business.



It is important to remember that this is estimated useful life then during its useful life it provides Incremental cash inflows but in the last year of the operation of that asset is the cash inflow or may be the cash outflow but it is called the **Life End Cashflow or Terminal Cashflow.**

It means that we name the Terminal Cashflow as the Cashflow of last year. It is inflow but it may be outflow also.

Terminal Cash Inflows		
a		Additional Revenue
b	-	Additional Depreciation
c	=	Net Change in Profit before Tax
d	-	Additional Income Tax
e	=	Net Change in Profit after Tax
f	+	Additional Depreciation
g	=	Incremental Cash Inflow of Last Year
h	+	Salvage Value of the Asset
i	-	Tax on Sale of the Asset
j	+	Release of Working Capital of the Project
k	=	Net Terminal Cashflow

Salvage value of asset: The Salvage value of an asset or residual value of asset or terminal value of asset is that value on which an asset can be sold at the end of its useful life. We may call it the **Scrap Value**, it is minimum but it is certainly a value which will bring cash inflow to us.

So, at the end of its useful life, when we will sell the asset for a certain amount then this will be our profit or gain and on this gain or profit we have to pay some tax.

Working capital of the Project: Whenever we start a new project we have to allocate some working capital for this. **Working Capital** is actually that amount which we require to run that project and we have already discussed about that in previous lectures also. But this is actually the cost require for raw material purchahse, payment of overheads, payment of labor cost and other expense.

So, once a project is terminated or completed then that working capital which we have invested in this project will also be released meaning that this amount will also come back to us. And that will also be a cash inflow and that is why we add it.

Another possibility is that at the end of the useful life, if we do not get any gain or any revenue from the sale of the old asset meaning that we have not get any Scrap value or Salvage value of asset but we have to incur further cost to uninstall or dismantle or scrap that asset or project and this may also be a situation.

Terminal Cash Inflows		
a		Additional Revenue
b	-	Additional Depreciation
c	=	Net Change in Profit before Tax
d	-	Additional Income Tax
e	=	Net Change in Profit after Tax
f	+	Additional Depreciation
g	=	Incremental Cash Inflow of Last Year
h	-	Cost to Dispose of Asset
i	+	Tax Saving on Disposal Cost
j	-	Increase in Net Working Capital
k	=	Net Terminal Cashflow

If we have to incur additional cost to dismantle or dispose off the asset then it will be a cash outflow and therefore we deduct it from our final year's Incremental Cash inflow.

We can conclude our discussion that **Terminal Cashflow** is the sum of the last year's Incremental Cash inflow and the adjustments related to the disposal of the asset, tax adjustments and working capital.

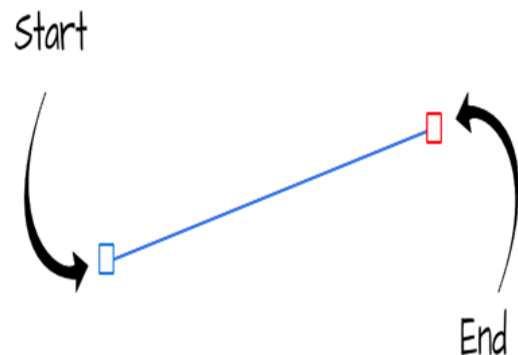
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Topic 171: Cash flows in Asset Replacement Decisions

This lecture is about **Capital Budgeting** and in this we will be talking about the Cashflows related to an investment in a project where we are replacing a project with a new one meaning that one project was already operational with us and its useful life has been completed and now we want to replace existing project with a new project.

So, **Replacement Decision** is actually a combination of two decisions: we have to salvage the previous asset or the previous project and we have to acquire the new asset or the new project.

- Terminal Cashflow of Old Asset
- Initial Cashflow of New Asset



We have discussed that we have cash outflow in doing investment in any new project that is when we are purchasing a new asset which requires a cash outflow and we have discussed in the previous lectures that how we calculate the Initial cash outflow which includes purchase price, direct cost that we will add up and calculate the Initial Cash outflow.

Second thing is that when we have to dispose off the old asset then we name the figure of Cashflow that we get in the last year of the useful life, the **Terminal**

Cashflow. So, this **Terminal Cashflow** is of the project which is Salvage which we are finishing/disposing and in place of this project we are taking other project and replacing it.

So, at the same time, in the same year, we have got the terminal Cashflow of the old project or the old asset and we also have the Initial Cashflow of the new asset. We have seen that usually the Terminal Cashflow is in the form of inflow because a big chunk of Incremental cash inflow is included in it and we do some adjustments (salvage value, tax, working capital,) in it also. But the **Initial cash outflow** which is for the purchase of new asset that is replacing asset, is certainly our outflow.

So, in the replacement decision on one hand, we have got inflow and on the other hand we have got outflow. So, the net value of that will be considered for the purpose of making a decision related to capital budgeting involving a decision related to replacement of an asset.

Cashflow of Replacement of Assets		
a		Cost of New Asset (Outflow)
b	+	Capitalized Expenditure of New Asset
c	+	Taxes and Duties
d	+	Installation and Other Costs
f	+/-	Cost to Dispose of Old Asset/ Proceeds of Sale of the Old Asset
g	-/+	Tax Saving on Disposal Cost/ Tax on Proceeds of Salvage Value
h	+/-	Increase/Decrease in Net Working Capital
i	=	Net Cashflow of Replaced Asset

Here, the '+' is representing the cash outflow because it shows that cost is increasing and when cost is increasing then obviously there is a cash outflow.

Here, the '-' is representing the cash inflow because it shows that cost is decreasing and when cost is decreasing then obviously there is a cash inflow.

We can conclude our discussion that in Replacement decision, two things are involved in it; one asset is involved which we are selling and terminating its operations and other asset is involved which we are purchasing new. We consider the cash inflows and cash outflows from these two assets' calculations. The Incremental Cashflow of old asset in the last year or terminal year which we do not consider it because it has no relation with new asset which we are replacing with that older one.

So, this calculation requires an Initial outflow of the new asset, the direct cost and the expenses plus any increase in the working capital and there are possibilities of additions or subtractions meaning outflows or inflows respectively in case of the cost of disposal or the revenue generated from the scrap value. In this way, we calculate the **cash flow in cash of replacement of an asset**.

Module 9: Capital Budgeting Techniques

Topic 172: Discounted Cashflow Methods-DCFs

In this lecture, we will discuss Discounted Cashflow Methods-DCFs which are actually used in Capital Budgeting. This is a continuation of our discussion on Capital Budgeting.

- Discounting/Present Value
- Discounted Cashflow Methods
 - Payback Period
 - IRR
 - NPV
 - PI



Discounting/Present Value:

In previous lectures, we have discussed about Time value of value where we said that **Present value** is the amount that we get by deducting interest from the some future amount that is given to us.

So, **Discounting** is a process of calculating the present value meaning that if any amount we have to get after 1 year or 2 year or 3 years then if we subtract the interest from that amount at a certain rate then we get the value which is the value of today and called the Present Value. So, **Discounted value** is the present value and **Discounting** is the process of calculating the present value.

Discounted Cashflow Methods:

When we say Discounted Cashflow Methods then it simply means those methods where we calculate the present values of the future Cashflows and from those present values, we come up with a certain decision related to the acceptance or rejection of a certain project.

It means that we use Discounted Cashflow methods to decide that should we invest in any project or not. It is obvious that its basis is on that this project is profitable or not or its profitability is that we want to get. So, this is something we determine by applying these Discounted Cashflow Methods where the objective is by using the present values, we come up to certain conclusions.

These Discounted Cashflow methods are commonly four in number:

Payback Period Method: In this method, we calculate that in how much time or years we will get our original investment which we have made in the project. It means that if we are putting \$1million then in how many years we will recover those \$1 million from that project.

So, this amount is calculated by considering the cash outflow and cash inflow meaning that if our investment in the project is \$1 million then how much inflows will be needed to recover that \$1million and in how many years those inflows will be received.

If that amount will be received in 5 years then we would say that the Payback period is 5 years. Then we compare the this Payback Period of project from our view with our Required Payback Period and make decision that whether the project is acceptable or not.

IRR (Internal Rate of Return) Method: In this method we calculate the rate of return of project meaning that this project will give us how much percent of return. And once we calculate that return we call it Internal Rate of Return (IRR) of the project.

Then we compare this Internal Rate of Return (IRR) with our Required Rate of Return meaning that we want that this project will give us 7% return then is project giving us 7% return and if it is giving this required rate then it is acceptable and if not giving us that 7% return then we will reject it.

Net Present Value (NPV): The present value is the value of the future amount or the future Cashflow today that is called **Present value**. So, if we deduct the outflow from the inflows meaning that on one side we have Initial Cash Outflow and on the other side we have Incremental Cash Inflow and if we find the present value of Incremental cash inflow by adding them and then deduct the Initial cash outflow from that figure then the value which we get is called the **Net Present Value (NPV)**.

If the present value of our inflows is more and outflow was less then this **Net Present Value (NPV)** will be positive, it will be some absolute amount 20,000, 30,000 or whatever it is. But if the present value of our inflows of our project is less than our outflows then we may get negative net present value. So, we have these possibilities of having a positive or negative and in very rare cases even a zero net present value.

Profitability Index (PI): In some ways, this method is very similar to Net Present Value (NPV). We would say similar because in Net Present Value (NPV) we take the difference of the inflows and outflows and in Profitability Index (PI) actually takes the ratio of the inflow and outflow.

In both Profitability Index (PI) and Net Present Value (NPV) methods, we calculate present value of inflows and outflow but in Net Present Value (NPV) we

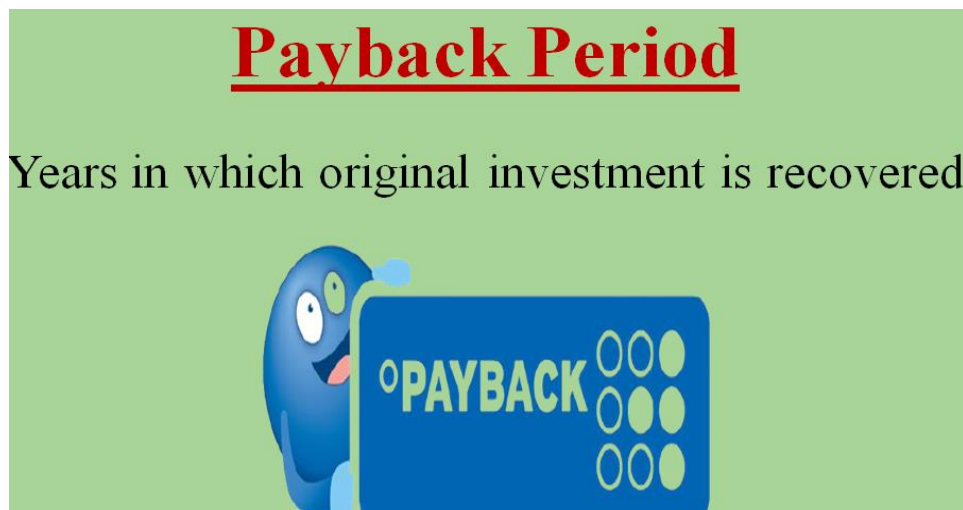
take difference means we subtract inflows and outflows but in Profitability Index (PI) method, we divide the amount of the present value of inflows with the outflows and determine an amount in the form of ratio or index.

So, we can conclude our discussion that Discounted Cashflow methods are those methods in which we use the present values of the future Cashflows and on the basis of those present values by relating them, adding them, subtracting them we come up with certain results which actually help us in deciding whether a certain project or a certain capital expenditure is acceptable to us or not.

Module 9: Capital Budgeting Techniques

Topic 173: Payback Period Method-1

In this lecture, we are starting our discussion on a Discounted Cashflow methods and the first one is the Payback Period.



Payback simply means that in how many years or in what time period, we will get our money back, we will get our investment back, the cost we are incurring in purchasing the asset, acquiring the asset will be returned to us from that project.

The **Payback Period** is the number of years in which our original investment, the capital expenditure will be recovered by generating the cash inflows.

We as a finance manager, a business man, a decision maker, know that this project should payback our amount in how many years. So, we calculate the Payback of the project and then we compare it with the required or targeted Payback Period.

Payback Period Method		
Year	Cash Flow	Accumulated Cashflow
	Rs.	Rs.
0	CASH OUTFLOW (100,000)	
1	20,000	20,000
2	40,000	60,000
3	40,000	100,000
4	50,000	140,000
5	60,000	200,000

Formula to Calculate the Payback Period:

$$\begin{array}{c}
 \text{Payback Period} \\
 \text{Formula}
 \end{array}
 = \frac{
 \begin{array}{c}
 \text{Initial Investment OR} \\
 \text{Original Cost of the Asset}
 \end{array}
 }{
 \begin{array}{c}
 \text{Cash Inflows}
 \end{array}
 }$$

How do we make a decision on the findings of Payback period whether the project is acceptable to us or not?

Although in this Payback Period method, there are some flaws which we will discuss in the next lectures. But here we will say that if we have calculated the payback period of a project then we compare that payback period with our required payback period.

There are three possibilities which may be considered as follows:

- If our payback period becomes equal to the required or actual payback period of the project then we will accept that project.
- If required or actual payback period is more than our payback period of that project then we will reject that project.
- And third possibility is that if this project has to recover our investment before our required payback period then it is very good than our expectations and we will also accept that project.

We can conclude our discussion that in order to calculate the Payback period; we have to calculate the Cashflows, we have to sum them up or accumulate them then we see that after how many years our amount is recovering and that is the Payback of the project.

Then we compare it with the required payback period and if the payback period of project is equal to or less than our required payback period then we accept that project. And if the payback period of project is more than our required payback period then we will reject that project and this is how we use the Payback Period Method.

Module 9: Capital Budgeting Techniques

Topic 174: Payback Period Method-2

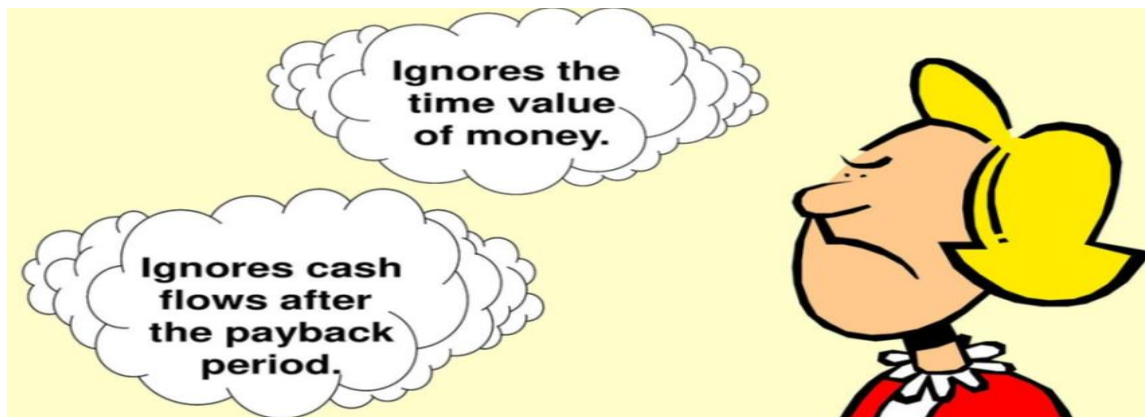
In first part of our discussion about Payback Period method, we have seen that how this Payback Period is calculated and how we make a decision on the basis of the results of Payback Period.

We have said that if the payback period of project is equal to or less than our required payback period then we accept that project. And if the payback period of project is more than our required payback period then we will reject that project and this is how we use the Payback Period Method.

The first part of this discussion of Discounted Cashflow Methods which we said that these methods are called DCFs or Discounted Cashflow Methods and where Discounting means to calculate present values.

But if we see that calculations of Payback Period in which we accumulate the Cashflows and we see that we have not discounted or have not calculated present values of these Cashflows.

So, this Payback Period has some flaws and the two basic flaws or two basic issues or problems this Payback Period are:



In its original form, Payback Period ignores time value of money. As Payback Period ignores the time value of money so, it cannot be called a Discounted Cashflow method.

This other problem with the Payback Period is that this method only is concern to the time period of recovery of the initial outflow. It means that this method does not consider the cash inflows coming after its payback period.

It may be that any project's Cashflows finished in 5th year and this 5th year is also its payback period. And it may be that any project's inflows continues till 5 to 10 or 15 years and there has been happen further cash inflow from that project but the Payback Period does not account for that.

So, the second flaw is that it ignores Cashflow after the payback period.

Payback Period Method

Payback Period Method using Present Value

Year	Cash Flow	PVs of Cash Flow	Accumulated Cashflow
	Rs.	@ 10% Rs.	Rs.
0	(100,000)	(100,000)	
1	20,000	18,182	18,182
2	30,000	24,793	42,975
3	40,000	30,053	73,028
4	50,000	34,151	107,179
5	60,000	37,255	144,434

Comparison of Two Projects				
Year	Project-A		Project-B	
	Cashflow	Accumulated Cashflow	Cashflow	Accumulated Cashflow
	Rs.	Rs.	Rs.	Rs.
0	(100,000)		(100,000)	
1	20,000	20,000	20,000	20,000
2	30,000	60,000	30,000	60,000
3	40,000	100,000	30,000	90,000
4	50,000	150,000	60,000	150,000
5	60,000	210,000	60,000	210,000
6			60,000	270,000
7			60,000	330,000
8			50,000	380,000

We can conclude our discussion that Payback Period is not originally the Discounted Cashflow method but by taking its present values we can make it Discounted Cashflows-DCF's but its second flaw that it does not consider the cash inflows after its Payback Period cannot be removed. That is the reason why this method is not considered a preferred method of making capital investment decisions.

Module 9: Capital Budgeting Techniques

Topic 175: Net Present Value-1

Net Present Value (NPV) is one of the Discounted Cashflow Method which is commonly used in Capital Budgeting.

Present value is the amount which is the value of a future amount today and it is calculated by deducting the interest or the return element which is there in the future amount, so when we deduct the interest or return according to the discounting rate from that future cash inflow then **we get Present Value** of that future cash inflow.

Net Present Value (NPV) means the difference between cash inflows and cash outflows. If we use the term **Net Cashflow** then it means that are inflows more or outflows. And there will be **Net cash inflow** then it means that cash inflow is more than cash outflow and if there will be Net cash outflow then it means that cash outflow is more than cash inflows.

But **in Net Present Value (NPV) method**, we are not only netting off the Cashflow but we are netting off the Present values of the Cashflows. The process is same as we deduct the cash outflow from the cash inflows but present values of the inflows and outflows.

This is not only the definition of Net Present Value (NPV) but this is also the method to calculate Net Present Value (NPV) meaning that **Net Present Value (NPV)** is the difference between the sum of Present Values of cash inflow and cash outflows.

This Net Present Value (NPV) can be positive or negative or it can be zero.



Formula to calculate Net Present Value (NPV):

Net Present Value		
Year	Cash Flow	PVs of Cash Flow
	Rs.	@ 10% Rs.
0	(100,000)	(100,000)
1	20,000	18,182
2	30,000	24,793
3	40,000	30,053
4	50,000	34,151
5	60,000	37,255
		144,434

$$\text{NPV} = 144,434 - 100,000 = 44,434$$

In this table, why are we using this 10% rate and why we are not using 20% or 30% and so on?

In technical terms, this is called the Discounting rate which is a rate used to calculate the present value but practically speaking this is our required rate of return meaning that when we are going to invest in any project then how much percent of profit or return we want to earn from that project and that is the rate which we use for the purpose of discounting the Cashflows.

So, the required rate may be the market rate or a predetermined rate or the opportunity cost.

In above table, the Net Present Value (NPV) is positive but it may be negative or zero. We can use positive and negative and zero values of Net Present Value (NPV) to make a decision on the basis of NPV meaning that when we have to accept a project. When the project is on, it is in two conditions or two possibilities;

- If the Net Present Value (NPV) is positive then we accept the project.

- Or If the Net Present Value (NPV) is 0 then we still accept the project.
- We only reject the project when the Net Present Value (NPV) is negative.

We can conclude our discussion on very important point that **why a project is still accepted at zero Net Present Value (NPV)?**

Its answer is that zero Net Present Value (NPV) means that the discounting rate or the required rate that we have used to calculate the Net Present Value (NPV) is actually the rate of the return of project meaning that the project is providing such rate of return which we want to earn from that project. And in this case, our Net Present Value (NPV) will become zero.

Because we have exactly used the same rate for discounting. And if any project is giving us exactly the same rate that we want to earn then definitely we will accept that project.

So, the acceptance is a positive or zero Net Present Value (NPV) and rejection is only if the Net Present Value (NPV) is negative. The last thing that we need to remember is that it is one method which gives us results in absolute terms meaning that this method gives us results in rupees or dollar value which we will further explore in our next lectures.

Module 9: Capital Budgeting Techniques

Topic 176: Net Present Value-2

As we have discussed that Net Present Value (NPV) is the amount that we get in access of our outflow of cash when we apply a certain discounting rate.

The discounting rate that we use to calculate the Net Present Value (NPV) is our Required Rate of Return which means that if we want to earn a certain percentage of return from a project then we have to discount the future cash inflows that will be generated by that project using that discounting rate. And when we sum up all those present values of future cash inflows that will be generated by the project and then subtract the outflow out of it and we get the Net Present Value (NPV).

If the Net Present Value (NPV) is a positive value that means that the discounting rate has still left as left a certain amount of profit in those Cashflows.

It means that if we have discounted all the cash inflows which we will have to receive in future on 20% and if we get a positive amount of Net Present Value (NPV) (I.e., 20,000, 80,000, 70,000 etc.) then it means that there was more than 20% profit included in those cash inflows. And when we discounted these cash inflows for 20% , we did not actually eliminate all the amount of the profit and hence we get a positive Net Present Value (NPV).

That is why we say that when Net Present Value (NPV) of project is zero, still we accept that project and the **reason for accepting a project with a Net Present Value (NPV) of zero** is that a zero (0) NPV shows that the project is giving us exactly that rate of return which we want to earn from that project.

From this, we can conclude that the rate of discounting directly affects the Net Present Value (NPV) and as our **discounting rate** (we need to remember that in the calculations of Net Present Value (NPV), the discounting rate is actually our required rate of return of project) will increase then our Net Present Value (NPV) will decrease with this increase in discounting rate.

It means that:

The purpose of zero Net Present Value (NPV) is that our project is providing exactly that rate of return which we want to earn from that project.

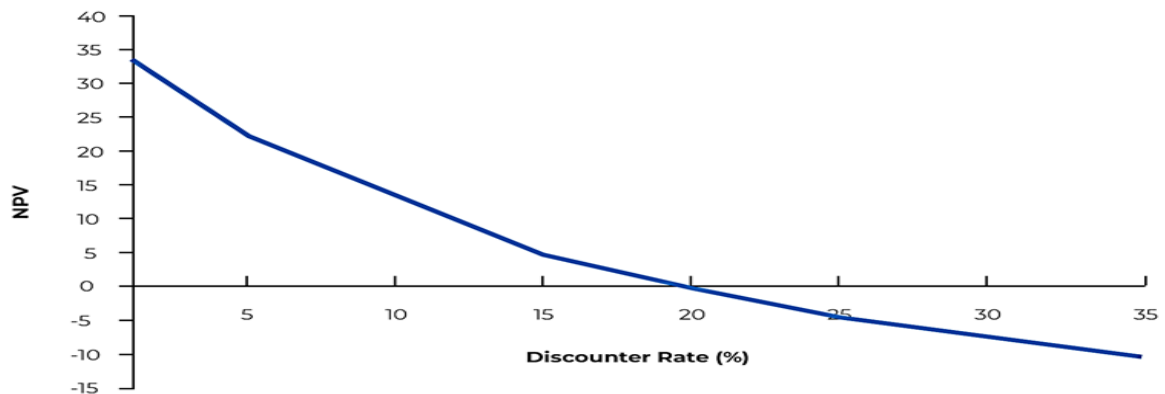
And the purpose of positive Net Present Value (NPV) is that the rate of return of that project is more than our required rate of return

And similarly, negative Net Present Value (NPV) means that rate of return of that project is less than our required rate of return.

That is why we have got the acceptance criteria is at zero or positive Net Present Value (NPV) and the **rejection criteria** is only that if Net Present Value (NPV) is negative.



Example of an NPV Profile



We can see that as our required rate or discounting rate will be increasing then we can see that our Net Present Value (NPV) is decreasing with this increase in discounting rate.

This graph tells us that the increase of discounting rate or required rate decreases our Net Present Value (NPV). And we can see that at 20% discounting rate, our Net Present Value (NPV) is zero that is why we can say that Higher is the discounting rate, lesser is the Net Present Value (NPV).

**NPV and
Discount Rate**
Higher is the discount
rate lesser is the NPV



How Net Present Value (NPV) is calculated?

Net Present Value		
Sr. No.	Rate	NPV
	%	Rs.
1	5%	21,000
2	10%	11,000
3	15%	5,000
4	20%	0
5	25%	(5,000)

As we have discussed that rate of return or that discounting rate where our Net Present Value (NPV) becomes zero that shows the actual rate of return provided by the project and we also name it as **IRR (Internal Rate of Return)**.

But if we further increase the required rate of return, we move into the rejection region because our Net Present Value (NPV) becomes negative.

We can conclude our discussion that Net Present Value (NPV) profile tells us that how much we can increase our required rate of return or discounting rate and at what point our project's viability will be expired or finished.

By looking in above table or set of data, we can say that this project is viable only upto 20% required rate of return and anything above 20% will lead to the rejection of the project.

Module 9: Capital Budgeting Techniques

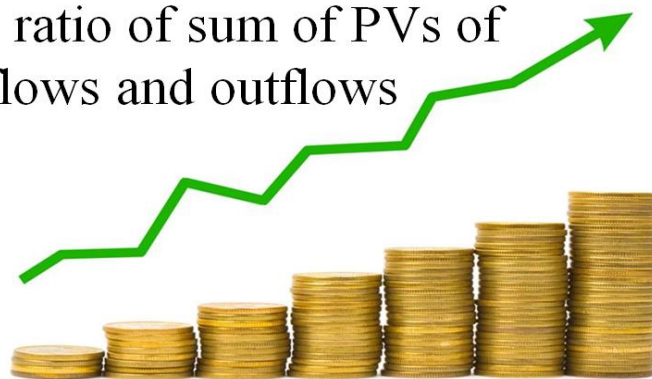
Topic 177: Profitability Index

Profitability Index (PI) is a part of our discussion on **Capital Budgeting**.

By definition, **Profitability Index (PI)** is a ratio between the sum of present values of cash inflows and cash outflows.

Profitability Index-PI

It is the ratio of sum of PVs of inflows and outflows



What is an Index?

The term **Index** is generally used for a relationship of two values which is mostly in the form of ratio and this Index is then used as a measure to determine or to assess something.

Now in this case, **Profitability Index (PI)** as the name also suggests is a tool or a method that we use to assess the projects for the purpose of making investments.

When we want to make investments in projects which require huge cash outflow, we use IRR (Internal Rate of Return) or NPV (Net Present Value) or PI (Profitability Index) or other methods. So, Profitability Index (PI) is one of those methods which give us very good point of view, a very good angle of assessing our returns of the project in terms of profitability.

Difference between NPV (Net Present Value) and PI (Profitability Index):

This is something which is also related to NPV (Net Present Value) meaning that we have seen in the discussion of **NPV (Net Present Value)** that these two things (sum of present values of inflows and outflows) which we relate. **In NPV (Net Present Value)**, we have taken difference of inflows and outflows meaning that we have subtracted the amount of outflows from the inflows and get NPV. So, NPV gives us a dollar or rupees value and we have certain criteria of acceptance and rejection.

In **PI (Profitability Index)**, instead of subtracting the outflows from the inflows, we actually develop a ratio that is we divide the inflows with the outflows and we come up with a certain value which may be equal to 1 or may be more or less than 1.

Profitability Index		
Year	Cash Flow	PVs of Cash Flow
	Rs.	@ 10% Rs.
0	(100,000)	(100,000)
1	20,000	18,182
2	30,000	24,793
3	40,000	30,053
4	50,000	34,151
5	60,000	37,255
		144,434

$$\text{PI} = 144,434/100,000 = 1.44$$

Once we have calculated the ratio or value of Index then the next or the most important thing is that to decide on the basis of the results that we have got whether to accept that project or not. Because the purpose of applying all these Discounted Cashflow methods is to decide whether we want to go for the project or not.

Acceptance and Rejection Criteria

If PI is less than 1
REJECT



So, we will decide to accept or reject any project on the basis whose simple criterion is that **if the value of Profitability Index (PI) is less than 1** then we will reject the project. Its result comes that **if the value of Profitability Index (PI) will be 1 or more than 1** then we will accept that project.

When the value of Profitability Index (PI) of any project is 1 or more than 1 then it means that this project is providing us our required rate return or providing rate of return more than our required rate of return.

So, any value which is greater than 1 indicates that the project is giving us a value of return which is more than our required rate of return and absolute 1 value shows that the project's rate of return is exactly equal to the required rate of return.

We can conclude our discussion of Profitability Index (PI) by stating that **Profitability Index (PI)** is a method of evaluating a project for investment purposes and it gives us an Index value which revolves around 1.0 and any value less than 1 is something that leads to the rejection of the project otherwise the project is accepted.

Module 9: Capital Budgeting Techniques

Topic 178: Internal Rate of Return-IRR Method-1

In previous lectures, we have talked about Discounted Cashflow Methods that what these actually are where we have said that every Discounted Cashflow method requires calculation of present values of Cashflows.

So, first we determine the Cashflows that will be generated by a project and then we calculate the present values of those Cashflows and that is why we call them the Discounted Cashflow methods.

Internal Rate of Return (IRR) is a method which actually may be called as a relative method of making a decision about a project.

What is relative and absolute and how we can distinguish both of them?

For this, here we are comparing it with Net Present Value (NPV). Internal Rate of Return (IRR) method is a method in which we get a certain rate of return which is in percentage that is; we will get the answer as 12% or 15% or 23%. Now a percentage is not an absolute amount.

If I calculate 10% of 1000 = $10/100 * 1000 = \text{Rs.}100$ but 5% of 20000 = $5/100 * 20000 = \text{Rs.}1000$ which is more than Rs.100. So, it means that a lower percentage may give me a bigger value if multiplied by an amount which is greater than the first value.

So, percentage gives us relative information and does not give information in the form of dollar or rupees. Net Present Value (NPV) is only method that gives us the results in the form of absolute dollar terms. Internal Rate of Return (IRR) is one method that gives us results in the form of percentage.

What does Internal Rate of Return (IRR) tell us?

Internal Rate of Return (IRR) tells us that how much percent of return this project will generate for us in which we want to make investment. When we apply this Internal Rate of Return (IRR) method, we get a result let's say 23% and then we compare this Internal Rate of Return (IRR) with our required rate of return. As

we said earlier that **required rate of return** is that rate which we want to earn from our project in which we are making investment.

And it is not essential for every investor that this required rate of return is same. There is a possibility that two investors may have got different requirements for earnings for a project and this is because of the opportunity cost of funds. If I have an investment opportunity which is more than 23% so, I would use a higher rate and if others do not have that opportunity so, they would probably be happy with a lower rate of return.

Technically speaking, **Internal Rate of Return (IRR)** with reference to the Cashflows is: we have to determine the present values of all the cash inflows and outflows then we have to see that the rate of return which actually equates or makes equal the sum of present values of cash inflows and outflows.

It means that if we keep the present value of cash outflows on one side of equation and the present values of cash inflows on other side of the equation then both side of that equation will become equal.

The rate of return for which the present values of cash inflows becomes equal to the present value of cash outflows, we name such rate of return the **Internal Rate of Return (IRR).**

How Internal Rate of Return (IRR) is calculated? Or how we can find that rate for which present values of inflows and present values of outflows becomes equal?

Its answer may be little bit un-scientific for some extent because the method that we use for it is called **Hit and Trial Method**. This Hit and Trial Method means that we apply different rates of return one after the other and try to calculate the present values of the Cashflow stream.

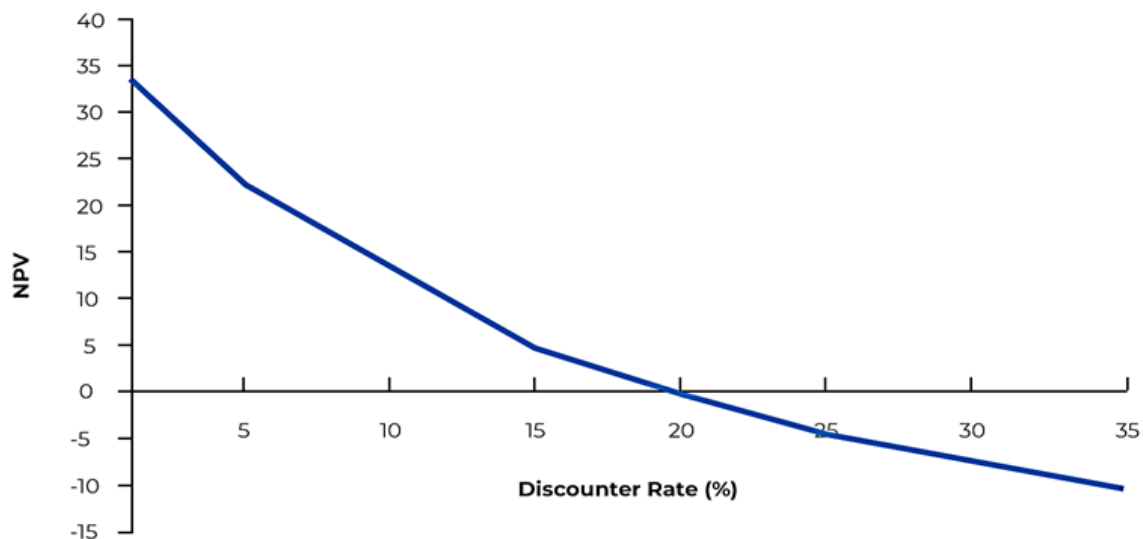
It means that we continue to apply different rates of return one after other and find present values of Cashflow streams and see that on which rate or return our inflows and outflows becomes equal.

If we are going to invest anywhere anyway we have an idea that how much rate of return we will get from that project. For example, if we are getting 18% rate of

return from the Stock Market or from the Real Estate market we can get 21% and in the business in which we are going generally it is generating lets say it is generating 25% so, this 25% then becomes the benchmark and we see by calculating our cash inflows and outflows near to that benchmark.



Example of an NPV Profile



Higher is the discounting rate, lesser is the Net Present Value (NPV) and then we see that there is a point which is like 20% where we have Net Present Value (NPV) becomes zero. This is that exact rate of return which that project is providing us. And this is that rate of return where sum of present values of inflows becomes equal to the sum of present values of outflows.

We can conclude our discussion that **Internal Rate of Return (IRR)** is a Discounted Cashflow method in which we determine present values and try to equate the present values of both inflows and outflows and then determine such rate on which both inflows and outflows becomes equal for which we use the Hit and Trial method meaning we continue to apply one rate after the other even that we reach at a point where both inflows and outflows become equal. So, this NPV profile tells us that how we move towards our Internal Rate of Return (IRR) by increasing rate of return.

Module 9: Capital Budgeting Techniques

Topic 179: Internal Rate of Return-IRR Method-2

In the last lecture, we have discussed that what theoretically IRR is and what kind of discounted Cashflow method IRR is and in this lecture, we will see that how the value of IRR, the percentage of IRR is calculated.

As we have talked about the process of calculating IRR and we said that the method or the process of calculating IRR is called a **Hit and Trial Method**.

Hit and Trial Method is that we assume a certain rate of return, apply that rate and see whether we get IRR or not. If we do not get then we move to another rate of return and then to other rate of return.

If we have applied any rate of return and we do not get the exact answer then what should we do? Should we increase rate of return or we decrease our rate of return?

Internal Rate of Return-IRR				
Year	Cash Flow	PVs of Cash Flow	PVs of Cash Flow	PVs of Cash Flow
	Rs.	@ 10% Rs.	@ 20% Rs.	@ 25% Rs.
0	(100,000)			
1	20,000	18,182	16,667	16,000
2	30,000	24,793	20,833	19,200
3	40,000	30,053	23,148	20,480
4	50,000	34,151	24,112	20,480
5	60,000	37,255	24,112	19,661
		144,434	108,872	95,821

We can see that if we discount this cash inflow at the 10% rate of return or discounting rate then the Net Present Value we will get or total Present value is 144,434 and we said that it should be equal to initial cash outflow meaning that if cash outflow is 100,000 then this Net Present Value should be equal to 100,000 whereas this is actually 144,434 which is more than our required amount of cash outflow.

As we know that if the discounting rate will more then our present value will be less.

So, what we need to do is that we basically need to increase the rate of return so that we can reduce the present value meaning that more will the discounting rate, less will our present value.

If we have got 144,434 amount of present value at 10% rate of return then we need to increase the discounting rate from 10% to any percentage.

Therefore we will move to other discounting rate which is 20% and we see that whether we get the present value equal to 100,000 or not. Again we see that still we are not getting 100,000 which is our initial cash outflow.

So, we further increase the discounting rate of return and we call this method **Hit and Trial Method** that we have used one rate of return and then we have increase that rate and then again increased or decreased that rate and even we reach at a point that for one rate of return, we have the value which is more than our outflow and for other rate of return, we have less value than our cash outflow.

This is that point where we have to know that our Internal Rate of Return (IRR) lie somewhere between 20% - 25%.

We can conclude our discussion that for the calculation of Internal Rate of Return (IRR), we have to apply rate of returns one after the other and if for which required rate of return we have calculated present values first time, and if the result becomes more than outflow then we have to increase discounting rare or rate of return.

And suppose if we got present value less than our cash outflow which is 100,000 then we will have to decrease this discounting rate or rate of return. it means that **if the present value is more than outflow** then **we will increase discounting rate**

or rate of return and if the present value is less than outflow then we will decrease discounting rate or rate of return.

And continue to apply difference discounting rate or rate or return till we get two such discounting rates in which on one rate our present value of cash inflow is more than outflow and on the other rate, our present value of cash inflow is less than our outflow.

Our Internal Rate of Return (IRR) will lie somewhere between these two values and that is 20%-25% in this case. It may be 21% or 23% or may be 24.5% and how we can determine that what is that exact rate for which we use the process called **Interpolation** that we will discuss in next lectures.

Module 9: Capital Budgeting Techniques

Topic 180: Interpolation

We will be talking about Internal Rate of Return (IRR) and in Internal Rate of Return (IRR) we will discuss that how the process of Interpolation takes place and how by using the method of Interpolation we reach to an exact or very close to the actual Internal Rate of Return (IRR).

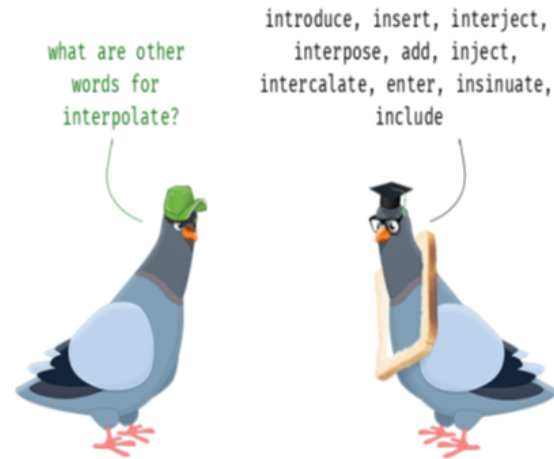
We have discussed that Internal Rate of Return (IRR) is the actual rate of return earned by the project for the investor. And to determine that Internal Rate of Return (IRR), we have to apply various rates of return one after the other and that is called the **Hit and Trial Method.**

If first time we have applied a rate of return and calculate present values then if sum of those present values becomes more than outflows then we have to move towards next higher rate of return. And if the sum of those present values becomes less than outflows then we have to reduce rate of return.

In this way, we move towards from one rate to the other rate even till we have got two such rates among which one rate is giving us higher value (present value of inflows becomes more than outflows) and other rate is giving lower value (present value of inflows becomes less than outflows).

Now we have to determine the exact value of Internal Rate of Return (IRR) among both these rates of return and for which we interpolate these two rates of return.

Interpolation
To determine an
unknown rate lying
between two known
rates



When we use the word 'Interpolation' in Internal Rate of Return (IRR) then it means that through interpolation we have to determine such an unknown rate of return between two given or known rates of returns which we cannot determine.

Internal Rate of Return-IRR				
Year	Cash Flow	PVs of Cash Flow	PVs of Cash Flow	PVs of Cash Flow
	Rs.	@ 10% Rs.	@ 20% Rs.	@ 25% Rs.
0	(100,000)			
1	20,000	18,182	16,667	16,000
2	30,000	24,793	20,833	19,200
3	40,000	30,053	23,148	20,480
4	50,000	34,151	24,112	20,480
5	60,000	37,255	24,112	19,661
		144,434	108,872	95,821

$$\text{IRR} = 20\% + 5\%[(108,872 - 100,000)/(108,872 - 95,821)]$$

Note: difference of selected rate of returns = 25% - 20% = 5%

$$= 0.2 + 0.05[8,872/13,051]$$

$$= 0.2 + 0.05[0.67979]$$

$$= 0.2 + 0.034 = 0.234$$

$$\text{IRR} = 0.234 \times 100 = 23.4\%$$

Here 23.4% is called Internal Rate of Return (IRR).

Is it truly the exact value of Internal Rate of Return (IRR)?

No, we can say that this may be or may not be the true value.

Then why it may be not the true value of Internal Rate of Return (IRR)?

When we have calculate the fraction we have rounded it off to the nearest decimal place that is: the answer of fraction is 0.33333.....but because we have rounded it off so, effect of rounding will appear here and the answer we get will not be 100% accurate.

Second thing is that whenever we interpolating meaning that difference between the two rates of returns that we interpolate will also make a result less accurate meaning that if we will interpolate the difference of 5% then it will be more accurate and if we interpolate the difference of 10% then it will be less accurate. And if we interpolate the difference of 2% then it will give us more precise result. So, that means; to determine the difference of 2%, then maybe we have to apply Hit and Trial method one or two times more.

Anyway this is the formula that we need to keep in our mind to calculate the value of Internal Rate of Return (IRR) using Interpolation.

On the basis of this result of Internal Rate of Return (IRR) after using interpolation, we will either accept the project or reject the project. We will compare this Internal Rate of Return (IRR) with our required rate of return which will be there before we actually start the calculation of Internal Rate of Return (IRR).

If Internal Rate of Return (IRR) is exactly equal to our required rate of return then we certainly accept the project.



If Internal Rate of Return (IRR) is more than our required rate of return, so it is even better that whatever we want to earn from the project, it is giving us even more than our required rate of return.

And finally if Internal Rate of Return (IRR) is less than our required rate of return meaning that the rate of return which we want to earn and that project is not providing such rate of return. And this becomes a basis of rejecting the project.

We can conclude our discussion that we will calculate Internal Rate of Return (IRR) by discounting the Cashflows by applying the Hit and Trial method then we

will interpolate those values and come up with a somewhat accurate value of the rate of return of the project.

And finally we compare it with the required rate of return. If Internal Rate of Return (IRR) will be equal or more than our required rate of return then project will be **accepted** and can make investment in it. And if Internal Rate of Return (IRR) will be less than our required rate of return then project will be **rejected**.

Module 9: Capital Budgeting Techniques

Topic 181: Capital Rationing-1

We will be talking about Capital Rationing in this lecture.

What is Capital Rationing or Rationing?

We have to do **Rationing** in every case where we have less available resources and their users are more than these resources. The process of Rationing is usually done in emergencies where our basic needs becomes less and people are more then we do not provide them things more than a specific quantity and we name it **Rationing**. When we are having a controlled distribution of some resources which are scarce or less then we call it **Rationing**.

Rationing

Controlled distribution of a scarce resource



Rationing is also done in **Capital Budgeting** meaning that rationing of investment. Any organization does not have unlimited funds available for investment and the funds which they have so, they have to invest these funds in more profitable way in given or available opportunities of investment.

It means that if we have 10 options available for investment then among which we have to select that option which will provide us more return or profit. If we can invest in these 10 opportunities then will go for all of them but if our investment resource is scarce or it is not enough to put in all those investments so, we will have to do Capital Rationing.

Capital Rationing

When profitable investment options are more than the funds available



When the profitable investment options are more than the funds available then we do Capital Rationing.

To do **Capital Rationing**, we have to select those projects which are all profitable meaning that such Discounted Cashflow methods (Net Present Value-NPV, Internal Rate of Return-IRR and Profitability Index-PI) which we have learnt

in Capital Budgeting then according to these three Discounted Cashflow methods, this project should be acceptable.

The criteria to be acceptable was that the value of Profitability Index should be 1 or more than 1, the value of NPV should be 0 or positive and IRR should be positive.

Project	Initial Cash Outflow	IRR %	NPV	PI
A	50,000	15	12,000	1.24
B	35,000	19	15,000	1.43
C	30,000	28	42,000	2.4
D	25,000	26	1,000	1.04
E	15,000	20	10,000	1.67
F	10,000	37	11,000	2.1
G	10,000	25	13,000	2.3
H	1,000	18	100	1.1

In this example, we are supposing that we have 65,000 rupees or dollars available investment. We can see that we cannot investment in all these projects from this Rs.65,000 so, we have to select some projects which are profitable or give us more return on investment.

We have given three criterion for selecting projects meaning that we have given results of IRR, NPV and PI value for each project.

We on the basis of all these Discounted Cashflow methods, firstly on the basis of IRR we will select such project whose IRR is more then such project whose IRR is less than the previous one and continue this process till our available investment Rs.65,000 get all consumed.

Then we will start selection on the basis of NPV then firstly we will select project whose NPV will be more then less than previous NPV value and so forth and continue this process till our available investment Rs.65,000 get all consumed.

Similarly, we will start selection on the basis of PI and we move from the project having more value of PI then will select the project who has less value of PI than previous one and continue this process till our available investment Rs.65,000 get all consumed.

Project	IRR	NPV	Outflow
F	37%	11,000	10,000
C	28%	42,000	30,000
D	26%	1,000	25,000
		54,000	65,000

Project	NPV	NPV	Outflow
C	42,000	42,000	30,000
B	15,000	15,000	35,000
		57,000	65,000

Project	PI	NPV	Outflow
C	2.4	42,000	30,000
G	2.3	13,000	10,000
F	2.1	11,000	10,000
E	1.67	10,000	15,000
		76,000	65,000

So, this is the process of Capital Rationing meaning that by considering three criteria (IRR, NPV and PI) one by one from available projects, we select the groups of projects on that basis of these three criterions and then calculate the value of NPV of every group.

We can conclude our discussion that **Capital Rationing** is such process through which we select such projects which can be more profitable for us. The basis of this selection will be IRR or NPV or the results of PI.

Module 9: Capital Budgeting Techniques

Topic 182: Capital Rationing-2

In our previous lecture, we have talked about what Rationing is, what **Capital Rationing** is and that is; when scarce resources are there and we need to make optimal use of that so, we go for the process of Rationing and if we have restricted available investment then we do Capital Rationing.

In previous lecture, we have seen that for doing Rationing we use one of the criteria among three criterions meaning that those **Discounted Cashflow methods** which we have discussed earlier (I.e., IRR, NPV, PI) and on the basis of one of them, the groups of projects which we select be considered as the best investment.

Our focus of discussion in this lecture with reference to the Capital Rationing will be **on the Decision Making**. We have done the process of Capital Rationing that how to select the group of projects from the many profitable available projects.

Decision Making in Capital Rationing

1. Maximum NPV
2. Why on the basis of NPV?
3. Why the combination selected on the basis of PI gives maximum NPV?



And now how can we decide that which group of projects is more profitable for us?

The answer is that combination of projects or that group of projects which gives us the maximum Net Present Value (NPV) meaning that the selected group of projects whose total Net Present Value (NPV) will be more than others then we will prefer these projects and make investment in them.

Why on the basis of Net Present Value (NPV) or make criteria, we will finally select the best or more profitable group of projects?

Or why Net Present Value (NPV) or its results are preferable and why the results of IRR and PI are not preferable?

The reason is that there values are in the form of absolute and relative terms meaning that in all these methods in which we take IRR, NPV and PI, then among these methods both IRR and PI are relative methods as they give us value in percentage or fraction and does not tell us that how much increase will be there in our total wealth (in the form of rupees or dollars etc.) if we make investment in these projects.

Net Present Value (NPV) is the only method that is an absolute method and it tells us precisely that in which combination of projects, there will be more increase in our total wealth and that is why we make basis Net Present Value (NPV) for making decision about our Capital Rationing.

Why the combination selected on the basis of PI gives maximum NPV?

The reason is that Profitability Index (PI) is actually represents a relationship between cash inflows and cash outflows meaning that the more of Profitability Index (PI) tells us that where we can earn more Net Present Value (NPV) with minimum investment.

So, we can conclude our discussion that **in Capital Rationing**, many groups are made on the basis of Discounted Cashflow methods and see that which group is giving us more Net Present Value (NPV) and we recommend that group of projects for investment. And the group of projects which actually always will give us the

maximum Net Present Value (NPV) will be the group or the combination that will be selected on the base of Profitability Index (PI).

Module 9: Capital Budgeting Techniques

Topic 183: Multiple IRRs (Internal Rate of Returns)

The Cashflows that we have considered in the Discounted Cashflow methods or in calculations of IRR and we take these usual or normal Cashflows meaning that first cash outflow has occur at the start of first year and second inflow occurs at the end of the first year and then similarly cash inflow is occurring at the end of every year.

This cash outflow at the start of first year and cash inflow at the end of every year is a theoretical assumption meaning that if we assume that we are going to invest in any larger project then it may be that one Cashflow occur at the start of first year and others in the middle and then at the end of the first year and it may be that cash outflow after the first year then continue to the second or third year also. But we keep it average or simplified in order to understand on the basic level.

In some projects, if cash outflows occur in first, second or third year or next years then there may be a possibility that we get value of IRR more than one. And we know that **IRR (Internal Rate of Return)** is that rate of return where the sum of present values of cash inflows becomes equal to the sum of present values of cash outflows.

And we have seen that where the Net Present Value (NPV) becomes zero and third thing we can say that where Profitability Index (PI) becomes exactly 1. So, **IRR (Internal Rate of Return)** is that rate of return for which NPV will be zero, PI will be 1.

This same relationship exist between NPV and PI that whenever NPV will be 0 then PI will always be 1. And if NPV will be more than 0 or positive then PI will

always be more than 1. And if NPV will be negative then PI will always be less than 1. In such a case where NPV is 0 and PI is 1 then it is our **IRR (Internal Rate of Return)**.

In some such situations where Cashflows exist and where we get more than one IRR (Internal Rate of Return) or get such rates more than one where our sum of present values of inflows and outflows becomes equal.

So, in some of these unusual cash flows we may end up with **multiple IRRs** (Internal Rate of Returns) or two IRRs for the same project.

Multiple IRRs

In some unusual cash flows we may end up with multiple IRRs i.e. two IRRs for one project.



Multiple IRRs			
Year	Cash Flows	PVs @ 25%	PVs @ 400%
0	-1600	-1600	-1600
1	10000	8000	2000
2	-10000	-6400	-400
		0	0

Here the confusion arises that sum of present values of both rates is 0 which means that our cash inflows and outflows become equal and both rates may be the Internal Rates of Return-IRRs but which rate can be selected.

The answer is that both rates are correct and the reason is that this is an unusual or impractical Cashflow.

We can conclude our discussion that **Multiple IRRs** is merely an academic discussion where we just explore the possibility of having more than one IRR because of the nature of the Cashflows. If we have such unusual Cashflows as we have seen above in which some outflows come after inflows then there may be a possibility that we may have more than one value of IRRs.

Module 9: Capital Budgeting Techniques

Topic 184: Sensitivity Analysis in Capital Budgeting

The term Sensitivity refers to the **Vulnerability**, the changes in the results or changes in the return with reference to various changes in the other variables which affect our return. It means that if there happens some changes in our actual investment or changes happen in our salvage value or estimation of inflow then what will be the impact of that change on the return.

That means if we are talking about accepting a certain project at a particular given outflow and a given certain inflows then if that outflow changes by a certain percentage then;

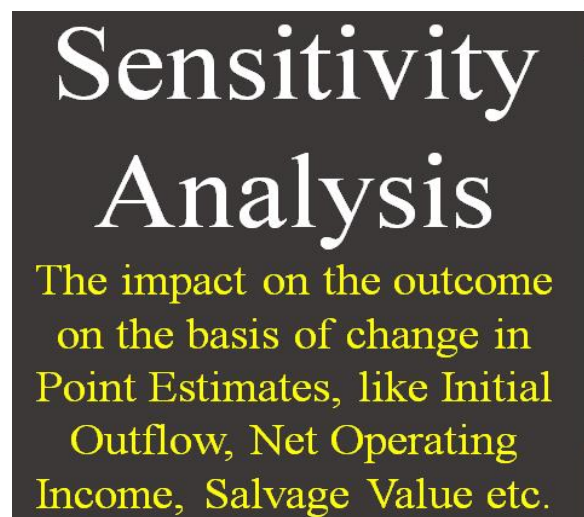
- Would that project still acceptable to us?
- Or if the outflow changes a bit then what did still we acceptable to us or not?

It means that with a little variation of inflows and outflows, if we apply our IRR (Internal Rate of Return), NPV (Net Present Value) and PI (Profitability Index) then changes happen in their results or not. And if changes happen by applying these Discounted Cashflow methods then there may be changes in preferences of project or not meaning that some projects will be properly accepted or some will

be down. This is what do we do in Sensitivity Analysis in Capital Budgeting.

Sensitivity Analysis in Capital Budgeting Definition:

It is something which is related to “The impact on the outcome or results on the basis of change in Point Estimates, like Initial Outflow, Net Operating Income or Incremental cash inflows, Salvage Value, Terminal Cashflow etc.” or how all these changes affect our investment decision and this is what we study in Sensitivity Analysis.



What do we do in this process of Sensitivity Analysis?

We actually take the initial or original set of data and that is the outflow and then incremental inflows and the terminal Cashflow.

Once we have done that then we have to see that if this is 15% or 20% or 30% and we will see that what changes would be there.

Sensitivity Analysis					
	-10%	-5%	Original Estimate	5%	10%
Initial Cash Outflow	90,000	95,000	Required cash outflow 100,000	105,000	110,000
Net Operating Income	22,500	23,750	Net operating income every year 25,000	26,250	27,500
Salvage Value	9,000	9,500	Terminal or scrap or residual value 10,000	10,500	11,000

Salvage value or scrap value or residual value is that value which is the value of an asset at the end of its useful life meaning that this value is less than its original value but by the way, it is such monetary value on which this asset can be disposed off.

So, **in Sensitivity Analysis**, we will actually see that if these values are fluctuate by 5 or 10 or 15 or 20% then what will be the change and finally what will be the impact on our results of discounted Cashflow methods (I.e., IRR, NPV, PI).

When we estimate Cashflows of project and that has done before the actual execution of project i.e., 3 months or 6 months before making actual investment in the project and we have done all these calculations of Discounted Cashflow methods.

But within that period of time, like within these 3 or 6 months when we actually go for the project then there is a certainly a possibility that the prices may go up that the returns might increase or the salvage value goes up or down and that means our initial outflow and incremental and the terminal Cashflow may be a little different from what we have expected.

So, this is the provision that we will include; before that time or occurrence of that situation we keep on exploring this possibility that if actual prices become increased or decreased then what can be the impact of that prices' increase or

decrease on our decision. So, we have not done the calculations of Discounted Cashflow methods on only on one set of data but on four or five or seven etc. set of data or certain possible fluctuations.

By doing so, we are in very good position that if our actual results increase or decrease from our estimate value then we also know that we have to make investment or not in this project.

One thing that we should keep in mind that it is not necessary that there may be same changes in our all Cashflows (Initial outflow, incremental inflows and terminal Cashflows) and it is possible that our initial cash outflow increases 10% but our profitability or operating profit or incremental cash inflow does not change and our scrap or salvage or terminal value also does not change.

So, we can do this Sensitivity Analysis in any way in which all values may be increased or decreased from a certain percentage or by increasing or decreasing one or two values among them then after that we calculate the values of Discounted Cashflows (DCF's) and see that on our acceptance or rejection criteria, this project after such these fluctuations remains same or there may be changes in our decisions. Or if our acceptance criteria converted into rejection with certain fluctuation then it means that we will have to change our decision after such variation.

So, this is a method which is very useful for making good decisions regarding capital investments.

Module 9: Capital Budgeting Techniques

Topic 185: Risk and Capital Budgeting

This is our concluding discussion on Capital Budgeting where we will talk about the Risk in Capital Budgeting.

What is Risk and how do we calculate Risk and how it is present in Capital Budgeting?

Risk:

Risk is not a negative thing as some of us consider it a negative thing. There are both positive and negative elements present in the Risk. Risk is actually an uncertainty or it is a possible variation from our expectations.

It means that when we say that we have to earn a certain given percentage of return and our actual return becomes more or less from that expected return then we will say that **there exist an element of Risk** therefore our return becomes more or less than our expected return.

So, the possible deviation from our expectation whatever we are expecting if we actually get more or less than that whatever we are expecting to invest, we have to put more or less. If our Cashflows expected were X and our actual Cashflows are X+ or X- something then we say that there is a level of Risk involve in that. And we show it with Uncertainty or Chances of Variation.

If we want to see by introducing Risk in Capital Budgeting that **how does it occur in Capital Budgeting** then we need to know that we get Cashflows from any business or project depends on which things.

If we say that any business or project operating in any country then it is a part of the economy of that country. If the economy is going up then business is also likely to go up and if the economy is going down then obviously that business will also be affected by that. So, if we consider various states or different possibilities of economy i.e., good, bad, moderate etc. and see that by using different tools or analysis methods that how many percent chances of this that this economy will be good.

If there has been given that how many chances for an economy to be good or bad or normal are there then by using them, then we can determine the value of uncertainty or risk occurring in our investment.

Economy	Probability	Cashflow
		Rs.
High Boom	0.1	10,000
Boom	0.2	9,000
Normal	0.3	7,000
Recession	0.3	5,000
High Recession	0.1	4,000

How do we calculate the uncertainty or the risk in these conditions with these Cashflows?

Formula to calculate uncertainty or risk of given Cashflows with specific states:

Average or Expected Cashflow = $CF \times P$

Measure of Risk or uncertainty/Standard Deviation = $(CF - Ave)^2 \times P$

Economy	Probability	Cashflow	CF x P	(CF-Ave) ² x P
		Rs.		
High Boom	0.1	10,000	1000	1,024,000
Boom	0.2	9,000	1,800	968,000
Normal	0.3	7,000	2,100	12,000
Recession	0.3	5,000	1,500	972,000
High Recession	0.1	4,000	400	784,000
	Expected or Average CF = 6,800			3,760,000
				SD = 1,939

Average or Expected Cashflow = $CF \times P$

Measure of Risk or uncertainty/Standard Deviation = **SD** = $(CF - Ave)^2 \times P$

Measure of Risk or uncertainty/Standard Deviation for High Boom

$$= (10,000 - 6,800)^2 \times 0.1$$

$$= (3,200)^2 \times 0.1$$

$$= 10,240,000 \times 0.1$$

$$= 1,024,000$$

$$\text{Variance} = 3,760,000$$

$$\text{SD} = \sqrt{3,760,000} = 1,939$$

It means that in above calculations, the element of Risk is 1,939. In other words, we can say that our expected Cashflow of 6,800 may increase or decrease from 1,939.

We can conclude our discussion that **Uncertainty or Risk** in Capital Budgeting is calculated by considering various states of economy where our economy's state is then according to that state we can adjust our Cashflows.

For that, it is necessary that firstly we determine the probability of every state of economy and then find the average Cashflow through the formula of mean or average and then calculate Standard deviation through the formula of standard deviation which will give us an element of Risk. This means that this is how risky our investment will be in these conditions of economy.